

- [About](#)
- [Online book](#)
- [Python Exercises](#)
- [Video Lectures](#)
- [Teaching Material](#)

17.2 Модель Хопфилда

Модель Хопфилда (226) состоит из сети N нейроны, обозначенные нижним индексом i где $1 \leq i \leq N$. Как и в некоторых более ранних моделях (335; 304; 549), нейроны в модели Хопфилда имеют только два состояния. Нейрон i активен, если его переменная состояния принимает значение $S_i = +1$ и «ВЫКЛ.» (без звука), если $S_i = -1$. Динамика развивается в дискретном времени с временными шагами Δt . Огнеупорность отсутствует, а продолжительность временного шага обычно не указывается. Если принять $\Delta t = 1$ мс, мы можем интерпретировать $S_i(t) = +1$ как потенциал действия нейрона i в момент времени t . Если взять $\Delta t = 500$ мс, $S_i(t) = +1$ скорее, это следует интерпретировать как эпизод с высокой частотой срабатывания.

Нейроны взаимодействуют друг с другом с помощью весовых коэффициентов w_{ij} . Входной потенциал нейрона i на которую влияет активность других нейронов

$$h_i(t) = \sum_j w_{ij} S_j(t). \quad (17.2)$$

Входной потенциал в момент времени t влияет на вероятностное обновление переменной состояния S_i на следующем временном шаге:

$$\text{Prob} \{S_i(t + \Delta t) = +1 | h_i(t)\} = g(h_i(t)) = g\left(\sum_j w_{ij} S_j(t)\right) \quad (17.3)$$

где g Это монотонно возрастающая функция усиления со значениями от нуля до единицы. Обычно используется $g(h) = 0.5 [1 + \tanh(\beta h)]$ с параметром β . Для $\beta \rightarrow \infty$, у нас есть $g(h) = 1$ для $h > 0$ и ноль в остальных случаях. Таким образом, динамика является детерминированной и описывается правилом обновления

$$S_i(t + \Delta t) = \text{sgn}[h_i(t)] \quad (17.4)$$

Для конечного β Динамика является стохастической. Далее мы предполагаем, что на каждом временном шаге все нейроны обновляются синхронно (параллельная динамика), но возможна и схема обновления, при которой за один временной шаг обновляется только один нейрон.

Цель этого раздела — показать, что при правильном выборе матрицы связи w_{ij} Элементы памяти могут быть восстановлены с помощью коллективной динамики, описанной в уравнении (17.3), применяемом ко всем N нейроны сети. Чтобы проиллюстрировать, как коллективная динамика может приводить к значимым результатам, в разделе 17.2.1 мы рассмотрим физику магнитных систем. В разделе 17.2.2 идеи, почерпнутые из физики магнитных систем, применяются к рассматриваемому вопросу — восстановлению воспоминаний.

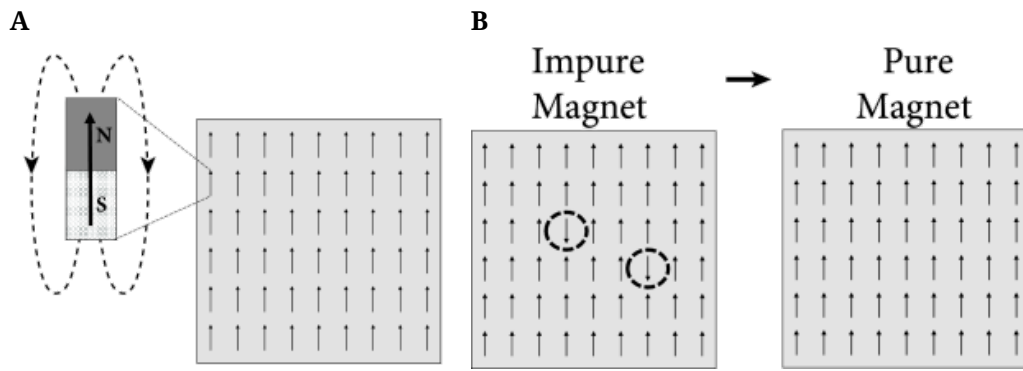


Рис. 17.5: Физика ферромагнетиков. **А.** Магнитные материалы состоят из атомов, каждый из которых обладает небольшим магнитным моментом, который здесь представлен в виде стрелки — символа магнитной стрелки. При низкой температуре все магнитные стрелки выровнены. Врезка: силовые линии вокруг одной из магнитных стрелок. **В.** При высокой температуре некоторые стрелки смещены (пунктирные круги). Охлаждение магнита приводит к самопроизвольной ориентации стрелок и восстановлению чистого магнита. Схематическое изображение.

17.2.1 Обходной путь: магнитная аналогия

Магнитный материал содержит атомы, обладающие так называемым спином. Спин создает магнитный момент на микроскопическом уровне, который графически изображается в виде стрелки (рис. 17.5А). При высокой температуре магнитные моменты отдельных атомов направлены во все возможные стороны. Однако при температуре ниже критической магнитные моменты всех атомов спонтанно выстраиваются в одном направлении. В результате микроскопические эффекты от всех атомных магнитных моментов суммируются, и материал приобретает макроскопические свойства ферромагнетика.

Чтобы понять, как может возникнуть самопроизвольное выравнивание, давайте изучим уравнения. (17.2) и (17.3) по аналогии с магнитными материалами. Мы предполагаем, что $w_{ij} = w_0 > 0$ между всеми парами нейронов $i \neq j$ и это самодействие исчезает, $w_{ii} = 0$.

Каждый атом характеризуется спиновой переменной $S_i = \pm 1$ где $S_i = +1$ указывает на то, что магнитный момент атома указывает «вверх». Предположим, что в момент времени $t = 0$ Все спины принимают положительное значение ($S_i = +1$), за исключением атома j которое имеет значение $S_j(0) = -1$ (Рис. 17.5А). Мы вычисляем вероятность того, что на временном шаге $t = \Delta t$, спин нейрона i переключится на $S_i = +1$ Эта вероятность определяется по формуле (17.3)

$$\text{Prob} \{S_i(t + \Delta t) = +1 | h_i(t)\} = g(h_i(t)) = g\left(\sum_{j=1}^N w_{ij} S_j(t)\right) = g(w_0 (N - 1)) \quad (17.5)$$

Здесь мы использовали наши допущения. При $g(h) = 0.5 [1 + \tanh(\beta h)]$ и $w_0 = \beta = 1$ Мы обнаружили, что для любой цепочки, состоящей более чем из трех атомов, вероятность того, что магнитные моменты всех атомов совпадут, чрезвычайно высока. В физических системах β играет роль обратной температуры. Если β При повышении температуры (высокая температура) магнитные моменты перестают выстраиваться в определенном направлении, и материал теряет спонтанную намагниченность.

Согласно уравнению (17.5), вероятность выравнивания увеличивается с ростом размера сети. Это особенность нашей модели, в которой все атомы взаимодействуют друг с другом. Однако физические взаимодействия быстро ослабевают с увеличением расстояния, поэтому сумма по j В уравнении (17.5) следует учитывать только ближайших соседей нейрона i Например, от 4 до 20 атомов в зависимости от конфигурации расположения атомов и дальности взаимодействия. Интересно, что нейроны, в отличие от атомов, способны к дальнему взаимодействию благодаря своим протяженным аксональным отросткам и дендритным деревьям. Таким образом, количество топологических соседей у каждого нейрона исчисляется тысячами.

Расположение идеально выровненных магнитных элементов выглядит довольно скучно, но в физике есть и более интересные примеры. В некоторых материалах, обычно состоящих из двух разных типов атомов, например А и В, возможно антиферромагнитное упорядочение (рис. 17.6). В то время как один слой магнитных моментов направлен вверх, следующий слой направлен вниз, так что макроскопическая намагниченность равна нулю. Тем не менее структура остается высокоупорядоченной. Примерами антиферромагнетиков являются некоторые оксиды и сплавы металлов.

Для моделирования антиферромагнетика мы выбираем взаимодействия $w_{ij} = +1$ если i и j принадлежат к одному классу (например, оба находятся в слое типа А или оба в слое типа В), и $w_{ij} = -1$ если один из двух атомов относится к типу А, а другой — к типу В. Простое повторение вычислений по формуле (17.5) показывает, что антиферромагнитная организация спинов возникает спонтанно при низкой температуре.

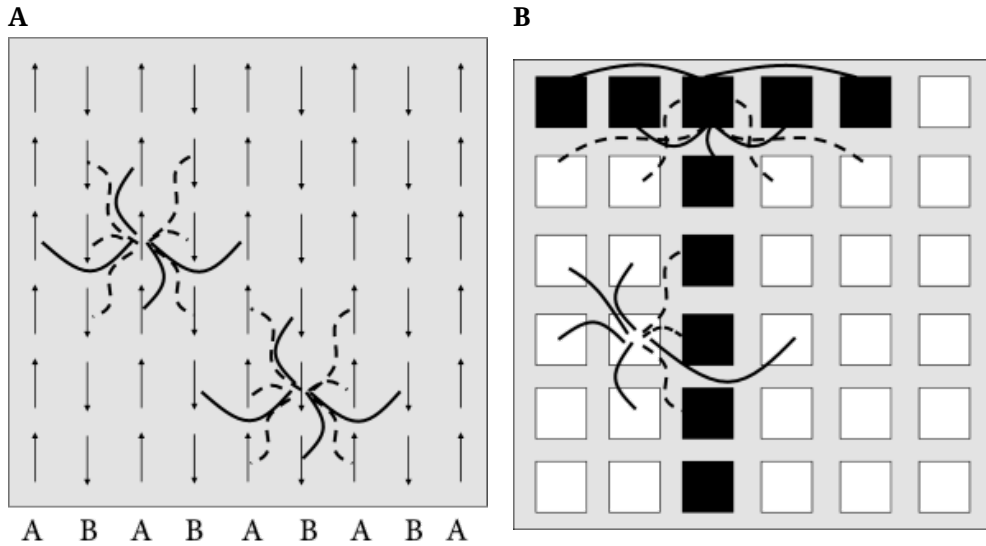


Fig. 17.6: Storing patterns. **A.** Physical anti-ferromagnets consist of layers of atoms A and B. All magnetic moments are aligned within a layer of identical neurons, but exhibit different orientations between layers. A model where interactions within atoms of the same type are positive (solid lines) and between atoms of different type are negative (dashed lines) can explain the spontaneous order in the arrangement of magnetic moments. The interaction scheme for two atoms with its 10 nearest neighbors is indicated. **B.** If we replace magnetic moments by black and white pixels (squares), represented by active and inactive neurons, respectively, the neuronal network can store a pattern, such as T. Interactions are positive (solid lines) between pixels of the same color (black-to-black or white-to-white) and negative otherwise. Only a few representative interactions are shown. Schematic figure.

The same idea of positive and negative interactions w_{ij} can be used to embed an arbitrary pattern into a network of neurons. Let us draw a pattern of black and white pixels corresponding to active ($p_i = +1$) and inactive ($p_i = -1$) neurons, respectively. The rule extracted from the anti-ferromagnet implies that pixels of opposite color are connected by negative weights, while pixels of the same color have connections with positive weight. This rule can be formalized as

$$w_{ij} = p_i p_j. \quad (17.6)$$

This rule forms the basis of the Hopfield model.

17.2.2 Patterns in the Hopfield model

The Hopfield model consists of a network of N binary neurons. A neuron i is characterized by its state $S_i = \pm 1$. The state variable is updated according to the dynamics defined in Eq. (17.3).

The task of the network is to store and recall M different patterns. Patterns are labeled by the index μ with $1 \leq \mu \leq M$. Each pattern μ is defined as a desired configuration $\{p_i^\mu = \pm 1; 1 \leq i \leq N\}$. The network of N neurons is said to correctly represent pattern μ , if the state of all neurons $1 \leq i \leq N$ is $S_i(t) = S_i(t + \Delta t) = p_i^\mu$. In other words, patterns must be fixed points of the dynamics (17.3).

For us as human observers, a meaningful pattern could, for example, be a configuration in form of a ‘T’, such as depicted in Fig. 17.6B. However, visually attractive patterns have large correlations between each other. Moreover, areas in the brain related to memory recall are situated far from the retinal input stage. Since the configuration of neurons in memory-related brain areas is probably very different from those at the retina, patterns in the Hopfield model are chosen as fixed random patterns; cf. Fig. 17.7.

During the set-up phase of the Hopfield network, a random number generator generates, for each pattern μ a string of N independent binary numbers $\{p_i^\mu = \pm 1; 1 \leq i \leq N\}$ with expectation value $\langle p_i^\mu \rangle = 0$. Strings of different patterns are independent. The weights are chosen as

$$w_{ij} = c \sum_{\mu=1}^M p_i^\mu p_j^\mu \quad (17.7)$$

with a positive constant $c > 0$. The network has full connectivity. Note that for a single pattern and $c = 1$, Eq. (17.7) is identical to the connections of the anti-ferromagnet, Eq. (17.6). For reasons that become clear later on, the standard choice of the constant c is $c = 1/N$.

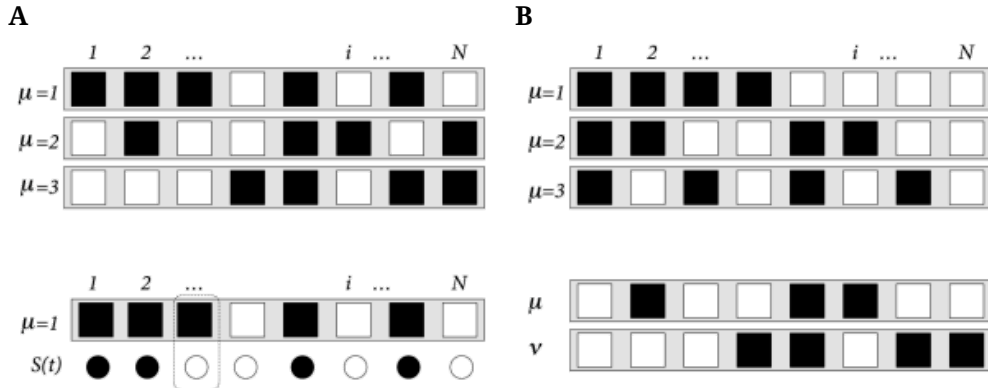


Fig. 17.7: Hopfield model. **A.** Top: Three random patterns $\mu = 1, 2, 3$ in a network of $N = 8$ neurons. Black squares ($p_i^\mu = +1$) and white squares ($p_i^\mu = -1$) are arranged in random order. Bottom: The overlap $m^1 = (1/N) \sum_i p_i^1 S_i(t)$ measures the similarity between the current state $S(t) = \{S_i(t); 1 \leq i \leq N\}$ and the first pattern. Here only a single neuron exhibits a mismatch (dotted line). The desired value in the pattern is shown as black and white squares, while the current state is indicated as black and white circles; schematic figure. **B.** Orthogonal patterns have a mutual overlap of zero so that correlations are $C^{\mu\nu} = (1/N) \sum_i p_i^\mu p_i^\nu = \delta^{\mu\nu}$ (top) whereas random patterns exhibit a small residual overlap for $\mu \neq \nu$ (bottom).

17.2.3 Pattern retrieval

In many memory retrieval experiments, a cue with partial information is given at the beginning of a recall trial. The retrieval of a memory item is verified by the completion of the missing information.

In order to mimic memory retrieval in the Hopfield model, an input is given by initializing the network in a state $S(t_0) = \{S_i(t_0); 1 \leq i \leq N\}$. After initialization, the network evolves freely under the dynamics (17.3). Ideally the dynamics should converge to a fixed point corresponding to the pattern μ which is most similar to the initial state.

In order to measure the similarity between the current state $S(t) = \{S_i(t); 1 \leq i \leq N\}$ and a pattern μ , we introduce the overlap (Fig. 17.7B)

$$m^\mu(t) = \frac{1}{N} \sum_i p_i^\mu S_i(t) . \quad (17.8)$$

The overlap takes a maximum value of 1, if $S_i(t) = p_i^\mu$, i.e., if the pattern is retrieved. It is close to zero if the current state has no correlation with pattern μ . The minimum value $m^\mu(t) = -1$ is achieved if each neuron takes the opposite value to that desired in pattern μ .

The overlap plays an important role in the analysis of the network dynamics. In fact, using Eq. (17.2) the input potential h_i of a neuron i is

$$h_i(t) = \sum_j w_{ij} S_j(t) = c \sum_{j=1}^N \sum_{\mu=1}^M p_i^\mu p_j^\mu S_j(t) = c N \sum_{\mu=1}^M p_i^\mu m^\mu(t) \quad (17.9)$$

where we have used Eqs. (17.7) and (17.8). In order to make the results of the calculation independent of the size of the network, it is standard to choose the factor $c = 1/N$, as mentioned above. In the following we always take $c = 1/N$ unless indicated otherwise. For an in-depth discussion, see the scaling arguments in Chapter 12.

In order to close the argument, we now use the input potential in the dynamics Eq. (17.3) and find

$$\text{Prob} \{S_i(t + \Delta t) = +1 | h_i(t)\} = g \left[\sum_{\mu=1}^M p_i^\mu m^\mu(t) \right] . \quad (17.10)$$

Eq. (17.10) highlights that the M macroscopic similarity values m^μ with $1 \leq \mu \leq M$ completely determine the dynamics of the network.

Example: Memory retrieval

Let us suppose that the initial state has a significant similarity with pattern $\mu = 3$, e.g., an overlap of $m^\mu(t_0) = 0.4$ and no overlap with the other patterns $m^\nu = 0$ for $\nu \neq 3$.

In the noiseless case Eq. (17.10) simplifies to

$$S_i(t_0 + \Delta t) = \text{sgn} \left[\sum_{\mu=1}^M p_i^\mu m^\mu \right] = \text{sgn} [p_i^3 m^3(t_0)] = p_i^3 \quad \text{for all } i \quad (17.11)$$

Hence, each neuron takes, after a single time step, the desired state corresponding to the pattern. In other words, the pattern with the strongest similarity to the input is retrieved, as it should be.

For stochastic neurons we find

$$\text{Prob} \{S_i(t_0 + \Delta t) = +1 | h_i(t)\} = g[p_i^3 m^3(t_0)] . \quad (17.12)$$

We note that, given the overlap $m^3(t_0)$, the right-hand side of Eq. (17.12) can only take two different values, corresponding to $p_i^3 = +1$ and $p_i^3 = -1$. Thus, all neurons that *should* be active in pattern 3 share the same probabilistic update rule

$$\text{Prob} \{S_i(t_0 + \Delta t) = +1 | h_i(t)\} = g[m^3(t_0)] \quad \text{for all } i \text{ with } p_i^3 = +1 \quad (17.13)$$

Similarly all those that *should* be inactive share another rule

$$\text{Prob} \{S_i(t_0 + \Delta t) = +1 | h_i(t)\} = g[-m^3(t_0)] \quad \text{for all } i \text{ with } p_i^3 = -1 \quad (17.14)$$

Thus, despite the fact that there are N neurons and M different patterns, during recall the network breaks up into two macroscopic populations: those that should be active and those that should be inactive. This is the reason why we can expect to arrive at macroscopic population equations, similar to those encountered in part III of the book.

Let us use this insight for the calculation of the overlap at time $t_0 + \Delta t$. We denote the size of the two populations by N_+^3 and N_-^3 , respectively, and find

$$\begin{aligned} m^3(t_0 + \Delta t) &= \frac{1}{N} \sum_i p_i^3 S_i(t_0 + \Delta t) \\ &= \frac{N_+^3}{N} \left[\frac{1}{N_+^3} \sum_{i \text{ with } p_i^3 = +1} S_i(t_0 + \Delta t) \right] - \frac{N_-^3}{N} \left[\frac{1}{N_-^3} \sum_{i \text{ with } p_i^3 = -1} S_i(t_0 + \Delta t) \right] \end{aligned} \quad (17.15)$$

We can interpret the two terms enclosed by the square brackets as the average activity of those neurons that should, or should not, be active, respectively. In the limit of a large network ($N \rightarrow \infty$) both groups are very large and of equal size $N_+^3 = N_-^3 = N/2$. Therefore, the averages inside the square brackets approach their expectation values. The technical term, used in the physics literature, is that the network dynamics are ‘self-averaging’. Hence, we can evaluate the square brackets with probabilities introduced in Eqs. (17.13) and (17.14). With $\text{Prob} \{S_i(t_0 + \Delta t) = -1 | h_i(t)\} = 1 - \text{Prob} \{S_i(t_0 + \Delta t) = +1 | h_i(t)\}$, we find

$$m^3(t_0 + \Delta t) = \frac{1}{2} \{2g[m^3(t_0)] - 1\} - \frac{1}{2} \{2g[-m^3(t_0)] - 1\} . \quad (17.16)$$

In the special case that $g(h) = 0.5 [1 + \tanh(\beta h)]$ Eq. (17.16) simplifies to an update law

In the special case that $g(x) = \tanh[\beta(1 + \tanh(\beta x))]$, Eq. (17.15) simplifies to an update law

$$m^3(t + \Delta t) = \tanh[\beta m^3(t)] \quad (17.17)$$

where we have replaced t_0 by t , in order to highlight that updates should be iterated over several time steps.

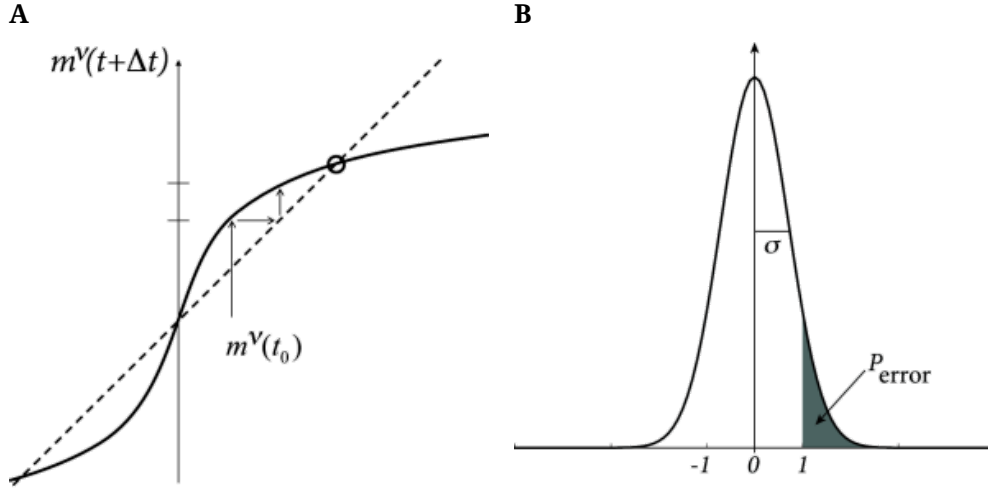


Fig. 17.8: Memory retrieval in the Hopfield model. **A.** The overlap $m^\nu(t + \Delta t)$ with a specific pattern ν is given as a function of the overlap with the same pattern $m^\nu(t)$ in the previous time step (solid line); cf. Eq. (17.16). The overlap with the $M - 1$ other patterns is supposed to vanish. The iterative update can be visualized as a path (arrow) between the overlap curve and the diagonal (dashed line). The dynamics approach a fixed point (circle) with high overlap corresponding to the retrieval of the pattern. **B.** The probability P_{error} that during retrieval an erroneous state-flip occurs corresponds to the shaded area under the curve; cf. Eq. (17.20). The width σ of the curve is proportional to the pattern load M/N ; schematic figure.

We close with three remarks. First, the dynamics of N neurons has been replaced, in a mathematically precise limit, by the iterative update of one single macroscopic variable, i.e., the overlap with one of the patterns. The result is reminiscent of the analysis of the macroscopic population dynamics performed in part III of the book. Indeed, the basic mathematical principles used for the equations of the population activity $A(t)$, are the same as the ones used here for the update of the overlap variable $m^\mu(t)$.

Second, if $\beta > 1$, the dynamics converges from an initially small overlap to a fixed point with a large overlap, close to one. The graphical solution of the update of pattern $\nu = 3$ (for which a nonzero overlap existed in the initial state) is shown in Fig. 17.8. Because the network dynamics is ‘attracted’ toward a stable fixed point characterized by a large overlap with one of the memorized patterns (Fig. 17.9A), the Hopfield model and variants of it are also called ‘attractor’ networks or ‘attractor memories’ (24; 40).

Finally, the assumption that, apart from pattern 3, all other patterns have an initial overlap exactly equal to zero is artificial. For random patterns, we expect a small overlap between arbitrary pairs of patterns. Thus, if the network is exactly in pattern 3 so that $m^3 = 1$, the other patterns have a small but finite overlap $|m^\mu| \neq 0$, because of spurious correlations $C^{\mu\nu} = (1/N) \sum_i p_i^\mu p_i^\nu$ between any two random patterns μ and ν ; Fig. 17.7B. If the number of patterns is large, the spurious correlations between the patterns can generate problems during memory retrieval, as we will see now.

17.2.4 Memory capacity

How many random patterns can be stored in a network of N neurons? Memory retrieval implies pattern completion, starting from a partial cue. An absolutely minimal condition for pattern

completion is that at least the dynamics should not move *away* from the pattern, if the initial cue is *identical* to the complete pattern (215). In other words, we require that a network with initial state $S_i(t_0) = p_i^\nu$ for $1 \leq i \leq N$ stays in pattern ν . Therefore pattern ν must be a fixed point under the dynamics.

We study a Hopfield network at zero temperature ($\beta = \infty$). We start the calculation as in Eq. 17.9 and insert $S_j(t_0) = p_j^\nu$. This yields

$$\begin{aligned} S_i(t_0 + \Delta t) &= \text{sgn} \left[\frac{1}{N} \sum_{j=1}^N \sum_{\mu=1}^M p_i^\mu p_j^\mu p_j^\nu \right] \\ &= \text{sgn} \left[p_i^\nu \left(\frac{1}{N} \sum_{j=1}^N p_j^\nu p_j^\nu \right) + \frac{1}{N} \sum_{\mu \neq \nu} \sum_j p_i^\mu p_j^\mu p_j^\nu \right], \end{aligned} \quad (17.18)$$

where we have separated the pattern ν from the other patterns. The factor in parenthesis on the right-hand side adds up to one and can therefore be dropped. We now multiply the second term on the right-hand side by a factor $1 = p_i^\nu p_i^\nu$. Finally, because $p_i^\nu = \pm 1$, a factor p_i^ν can be pulled out of the argument of the sign-function:

$$S_i(t_0 + \Delta t) = p_i^\nu \text{sgn} \left[1 + \frac{1}{N} \sum_j \sum_{\mu \neq \nu} p_i^\mu p_i^\nu p_j^\mu p_j^\nu \right] = p_i^\nu \text{sgn}[1 - a_{i\nu}]. \quad (17.19)$$

The desired fixed point exists only if $1 > a_{i\nu} = \frac{1}{N} \sum_j \sum_{\mu \neq \nu} p_i^\mu p_i^\nu p_j^\mu p_j^\nu$ for all neurons i . In other words, even if the network is initialized in perfect agreement with one of the patterns, it can happen that one or a few neurons flip their sign. The probability to move away from the pattern is equal to the probability of finding a value $a_{i\nu} > 0$ for one of the neurons i .

Because patterns are generated from independent random numbers $p_i^\mu = \pm 1$ with zero mean, the product $p_i^\mu p_i^\nu p_j^\mu p_j^\nu = \pm 1$ is also a binary random number with zero mean. Since the values p_i^μ are chosen independently for each neuron i and each pattern μ , the term $a_{i\nu}$ can be visualized as a random walk of $N(M-1)$ steps and step size $1/N$. For a large number of steps, the positive or negative walking distance can be approximated by a Gaussian distribution with zero mean and standard deviation $\sigma = \sqrt{(M-1)/N} \approx \sqrt{M/N}$ for $M \gg 1$. The probability that the activity state of neuron i erroneously flips is therefore proportional to

$$P_{\text{error}} = \frac{1}{\sqrt{2\pi}\sigma} \int_1^\infty e^{-\frac{x^2}{2\sigma^2}} dx \approx \frac{1}{2} \left[1 - \text{erf} \left(\sqrt{\frac{N}{2M}} \right) \right] \quad (17.20)$$

where we have introduced the error function

$$\text{erf}(x) = \frac{1}{\sqrt{\pi}} \int_0^x e^{-y^2} dy \quad (17.21)$$

The most important insight is that the probability of an erroneous state-flip increases with the ratio M/N . Formally, we can define the storage capacity C_{store} of a network as the maximal number M^{max} of patterns that a network of N neurons can retrieve

$$C_{\text{store}} = \frac{M^{\text{max}}}{N} = \frac{M^{\text{max}} N}{N^2}. \quad (17.22)$$

For the second equality sign we have multiplied both numerator and denominator by a common factor N which gives rise to the following interpretation. Since each pattern consists of N neurons (i.e., N binary numbers), the total number of bits that need to be stored at maximum capacity is $M^{\text{max}} N$. In the Hopfield model, patterns are stored by an appropriate choice of the synaptic

connections. The number of available synapses in a fully connected network is N^2 . Therefore, the storage capacity measures the number of bits stored per synapse.

Example: Erroneous bits

We can evaluate Eq. (17.20) for various choices of P_{error} . For example, if we accept an error probability of $P_{\text{error}} = 0.001$, we find a storage capacity of $C_{\text{store}} = 0.105$.

Hence, a network of 10'000 neurons is able of storing about 1'000 patterns with $P_{\text{error}} = 0.001$. Thus in each of the patterns, we expect that about 10 neurons exhibit erroneous activity. We emphasize that the above calculation focuses on the *first* iteration step only. If we start in the pattern, then about 10 neurons will flip their state in the first iteration. But these flips could in principle cause further neurons to flip in the second iteration and eventually initiate an avalanche of many other changes.

A more precise calculation shows that such an avalanche does not occur, if the number of stored pattern stays below a limit such that $C_{\text{store}} = 0.138$ (18; 22).

17.2.5 The energy picture

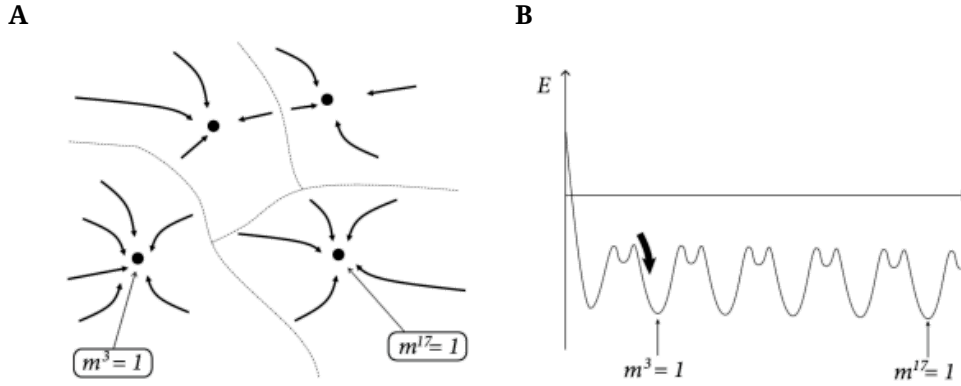


Fig. 17.9: Attractor picture and energy landscape. **A.** The dynamics are attracted toward fixed points corresponding to memory states (overlap $m^\nu = 1$). Four attractor states are indicated. The dashed lines show the boundaries of the basin of attraction of each memory. **B.** The Hopfield model has multiple equivalent energy minima, each one corresponding to the retrieval (overlap $m^\nu = 1$) of one pattern. Between the main minima, additional local minima (corresponding to mixtures of several patterns) may also exist.

The Hopfield model has symmetric interactions $w_{ij} = w_{ji} = c \sum_{\mu=1}^M p_i^\mu p_j^\mu$. We now show that, in any network with symmetric interactions and asynchronous deterministic dynamics

$$S_i(t + \Delta t) = \text{sgn}[h(t)] = \text{sgn} \left[\sum_j w_{ij} S_j(t) \right] \quad (17.23)$$

the energy

$$E = - \sum_i \sum_j w_{ij} S_i S_j \quad (17.24)$$

decreases with each state flip of a single neuron (226).

In each time step only one neuron is updated (asynchronous dynamics). Let us assume that after application of Eq. (17.23) neuron k has changed its value from S_k at time t to $S'_k = -S_k$ while all other neurons keep their value $S'_j = S_j$ for $j \neq k$. The prime indicates values evaluated at time $t + \Delta t$. The change in energy caused by the state flip of neuron k is

$$E' - E = - \sum_i w_{ik} S_i (S'_k - S_k) - \sum_j w_{kj} S_j (S'_k - S_k) . \quad (17.25)$$

First, because of the update of neuron k , we have $S'_k - S_k = 2S'_k$. Second, because of the symmetry $w_{ij} = w_{ji}$, the two terms on the right-hand side are identical, and $\sum_i w_{ik} S_i = \sum_i w_{ki} S_i = h_k$. Third, because of Eq. (17.23), the sign of h_k determines the new value S'_k of neuron k . Therefore the change in energy is $E' - E = -4h_k \operatorname{sgn} h_k < 0$. In other words, the energy E is a Liapunov function of the deterministic Hopfield network.

Since the dynamics leads to a decrease of the energy, we may wonder whether we can say something about the global or local minimum of the energy. If we insert the definition of the connection weights into the energy function (17.24), we find

$$E = - \sum_i \sum_j \left(c \sum_{\mu} p_i^{\mu} p_j^{\mu} \right) S_i S_j = -c N^2 \sum_{\mu} (m^{\mu})^2 , \quad (17.26)$$

where we have used the definition of the overlap; cf. Eq. (17.8).

The maximum value of the overlap with a fixed pattern ν is $m^{\nu} = 1$. Moreover, for random patterns, the correlations between patterns are small. Therefore, if $m^{\nu} = 1$ (i.e., recall of pattern ν) the overlap with other patterns $\mu \neq \nu$ is $m^{\mu} \approx 0$. Therefore, the energy landscape can be visualized with multiple minima of the same depth, each minimum corresponding to retrieval of one of the patterns (Fig. 17.9B).

17.2.6 Retrieval of low-activity patterns

There are numerous aspects in which the Hopfield model is rather far from biology. One of these is that in each memory pattern, fifty percent of the neurons are active.

To characterize patterns with a lower level of activity, let us introduce random variables $\xi_i^{\mu} \in \{0, 1\}$ for $1 \leq i \leq N$ and $1 \leq \mu \leq M$ with mean $\langle \xi_i^{\mu} \rangle = a$. For $a = 0.5$ and $p_i^{\mu} = 2\xi_i^{\mu} - 1$ we are back to the patterns in the Hopfield model. In the following we are, however, interested in patterns with an activity $a < 0.5$. To simplify some of the arguments below, we suppose that patterns are generated under the constraint $\sum_i \xi_i^{\mu} = N a$ for each μ , so that all patterns have *exactly* the same target activity a .

The weights in the Hopfield model of Eq. (17.7) are replaced by

$$w_{ij} = c' \sum_{\mu=1}^M (\xi_i^{\mu} - b) (\xi_j^{\mu} - a) \quad (17.27)$$

where a is the mean activity of the stored patterns, $0 \leq b \leq 1$ a constant, and $c' = [2a(1-a)N]^{-1}$. Note that Eq. (17.7) is a special case of Eq. (17.27) with $a = b = 0.5$ and $c' = 2c$.

As before, we work with binary neurons $S_i = \pm 1$ defined by the stochastic update rule in Eqs. (17.2) and (17.3). To analyze pattern retrieval we proceed analogously to Eq. (17.10). Introducing the overlap of low-activity patterns

$$m^\mu = \frac{1}{2a(1-a)N} \sum_j \left(\xi_j^\mu - a \right) S_j \quad (17.28)$$

we find

$$\text{Prob} \{S_i(t + \Delta t) = +1 | h_i(t)\} = g \left[\sum_{\mu=1}^M (\xi_i^\mu - b) m^\mu(t) \right]. \quad (17.29)$$

Example: Memory retrieval and attractor dynamics

Suppose that at time t the overlap with one of the patterns, say pattern 3, is significantly above zero while the overlap with all other patterns vanishes $m^\mu \approx m \delta^{\mu 3}$, where δ^{nm} denotes the Kronecker- δ . The initial overlap is $0.1 < m \leq 1$. Then the dynamics of the low-activity networks splits up into two groups of neurons, i.e., those that should be ‘ON’ in pattern 3 ($\xi_i^3 = 1$) and those that should be ‘OFF’ ($\xi_i^3 = 0$).

The size of both groups scales with N : There are $a \cdot N$ ‘ON’ neurons and $(1 - a) \cdot N$ ‘OFF’ neurons. For $N \rightarrow \infty$, the population activity A^{ON} of the ‘ON’ group (i.e., the fraction of neurons with state $S_i = +1$ in the ‘ON’ group) is therefore well described by its expectation value

$$A^{\text{ON}}(t + \Delta t) = g(1 - b) m^3(t). \quad (17.30)$$

Similarly, the ‘OFF’ group has activity

$$A^{\text{OFF}}(t + \Delta t) = g(-b) m^3(t). \quad (17.31)$$

To close the argument we determine the overlap at time $t + \Delta t$. Exploiting the split into two groups of size $a \cdot N$ and $(1 - a) \cdot N$, respectively, we have

$$\begin{aligned} m^3(t + \Delta t) &= \frac{1}{2a(1-a)N} \left[\sum_{j \text{ with } \xi_j^3=1} (1-a) S_j(t + \Delta t) + \sum_{j \text{ with } \xi_j^3=0} (-a) S_j(t + \Delta t) \right] \\ &= \frac{1}{2} [A^{\text{ON}}(t + \Delta t) - A^{\text{OFF}}(t + \Delta t)]. \end{aligned} \quad (17.32)$$

Thus, the overlap with pattern 3 has changed from the initial value $m^3(t) = m$ to a new value $m^3(t + \Delta t)$. Retrieval of memories works if iteration of Eqs. (17.30) - (17.32) makes m^3 converge to a value close to unity while, at the same time, the other overlaps m^ν (for $\nu \neq 3$) stay close to zero.

We emphasize that the analysis of the network dynamics presented here does not require symmetric weights but is possible for arbitrary values of the parameter b . However, a standard choice is $b = a$ which leads to symmetric weights and to a high memory capacity (522).